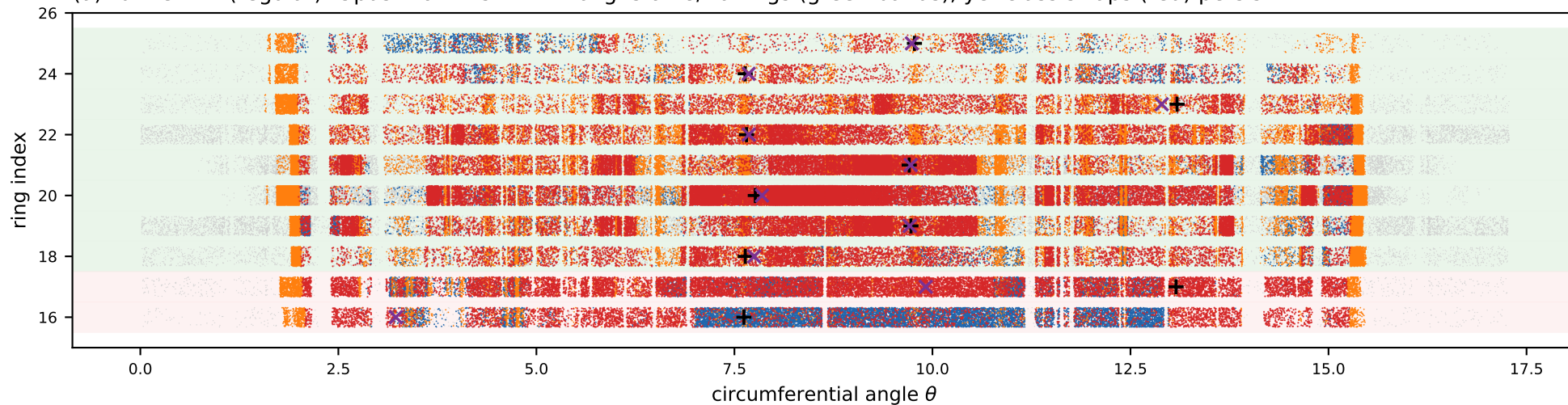
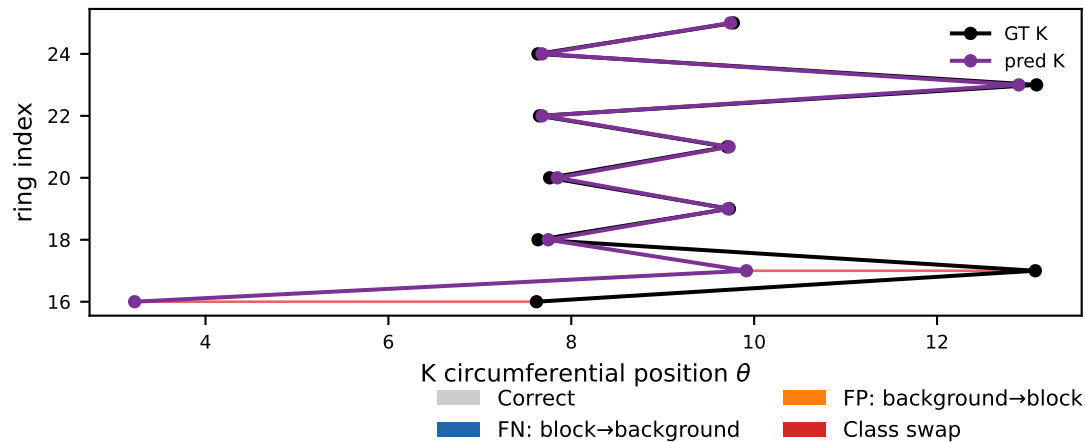


Regular tunnel: residual error is class swap despite a good K-anchor

(a) Tunnel 1-1 (regular): Opus-4.6 m+s+k — K aligns on 8/10 rings (green bands), yet class swaps (red) persist



(b) GT and predicted K track together on most rings (short connectors)



(c) Even K-aligned rings cap at 0.81 accuracy — swap 11% vs FN 1%

